ABSTRACT

In the Smart Personal Safety Wearable with Fall Detection, is a groundbreaking initiative aimed at addressing the growing concern of unnoticed falls among elderly individuals and patients. Falls often lead to severe injuries, and without timely intervention, they can escalate into critical health emergencies. Your wearable device is designed to provide an efficient, real-time solution by integrating high-accuracy motion detection sensors with Bluetooth-based alerts sent to caregivers through a mobile app. This instant communication ensures that assistance reaches the affected individual promptly, reducing response time and improving safety outcomes.

One of the key innovations of your project is its compact and lightweight design, making it comfortable for users to wear throughout the day without inconvenience. Additionally, the hardware components are optimized for low-cost production while maintaining high efficiency, ensuring affordability for a wide demographic. The feasibility and viability of the device extend beyond immediate safety benefits—it is adaptable, upgradable, and sustainable, allowing future enhancements that can integrate newer technologies such as AI-based predictive analytics for fall prevention.

Beyond individual benefits, your project has broader societal impacts. It fosters safety awareness among caregivers and family members while empowering elderly individuals to maintain their independence with greater confidence. By reducing fall-related hospitalizations and associated medical costs, it contributes to a more efficient healthcare system. The alignment of your project with Sustainable Development Goals (SDGs), particularly Goal 3 (Good Health and Well-being) and Goal 9 (Industry, Innovation, and Infrastructure), highlights its potential as a socially responsible and impactful solution.

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CHAPTER 1

INTRODUCTION

In today's rapidly aging society, elderly safety and independent living have become significant concerns. One of the most serious health threats for the elderly is falling. Falls can result in severe injuries, long-term disabilities, or even death if timely medical attention is not provided. In many cases, individuals are unable to call for help due to unconsciousness or immobility following a fall. This creates an urgent need for automated fall detection systems.

To tackle this issue, we have designed and developed a Smart Personal Safety Wearable equipped with real-time fall detection. This wearable device is compact, lightweight, and can be comfortably worn throughout the day. It uses an MPU6050 sensor to continuously monitor the user's movement and orientation. In the event of a sudden fall or abnormal motion pattern, the device immediately identifies the fall condition using a threshold-based algorithm.

The system is powered by an ESP32 microcontroller, which not only processes sensor data efficiently but also allows wireless connectivity. A buzzer provides an audible alert locally, while the OLED screen displays real-time status or emergency messages. The system is energy-efficient and can be powered through USB or a rechargeable battery, making it practical for everyday use.

Our goal with this project is to offer an affordable, open-source solution that enhances personal safety, especially for the elderly and individuals with health risks. The device can be expanded in the future to include location tracking, automatic emergency calling, and integration with healthcare monitoring platforms.

1.1 Objective:

The primary objective of this project is to design and develop a smart, wearable device capable of detecting accidental falls in real time. The system aims to provide enhanced safety and immediate alert mechanisms for elderly individuals and patients who are at a higher risk of falling. To achieve this, the project utilizes the MPU6050 sensor, which combines a 3-axis accelerometer and gyroscope to continuously monitor body movement and orientation. A threshold-based algorithm is implemented on an ESP32 microcontroller to identify fall conditions based on sudden changes in acceleration and orientation.

The device is designed to generate instant alerts through a buzzer and an OLED display, allowing nearby individuals to respond quickly in emergency situations. Portability, low power consumption, and ease of use are core design goals to ensure that the system is practical for everyday wear. Furthermore, the project emphasizes the use of open-source, low-cost components to maintain affordability. In the future, the device may be enhanced with wireless capabilities to send alerts to caregivers via Wi-Fi or Bluetooth, making it a scalable solution for personal safety monitoring.

1.2 Methodology:

The development of the Smart Personal Safety Wearable followed a structured approach, beginning with hardware selection and system design. The ESP32 microcontroller was chosen for its processing capability and built-in Wi-Fi/Bluetooth support. The MPU6050 sensor was integrated to provide continuous real-time monitoring of both linear acceleration and angular velocity across three axes. These sensor readings form the basis for detecting fall-like motion patterns.

The MPU6050 communicates with the ESP32 via the I2C protocol, ensuring efficient data transfer. An OLED display module is also connected using the same I2C bus to visually indicate system status and alerts. A 5V active buzzer is included to produce audible warnings when a fall is detected. The firmware is developed in the Arduino IDE using open-source libraries such as Adafruit_SSD1306, Adafruit_GFX, and MPU6050.

The core fall detection logic uses a threshold-based algorithm, which calculates the magnitude of the resultant acceleration vector from the MPU6050. If the acceleration value falls below a predefined limit (indicating free fall) and is followed by a spike in acceleration (impact), the system identifies it as a fall event. Upon detection, the buzzer is triggered and a warning is displayed on the screen. The entire system is powered via USB, with provisions for battery operation in future iterations.

1.2.1 Sensor Integration and Data Collection:

- The Smart Personal Safety Wearable system integrates the MPU6050 sensor, a 6-axis motion tracking device that combines a 3-axis accelerometer and a 3-axis gyroscope in a single module. This sensor is responsible for detecting

both linear acceleration and angular rotation, making it ideal for motion-based

- event detection such as falls. The MPU6050 is interfaced with the ESP32 microcontroller via the I2C communication protocol, which allows for efficient and reliable data exchange using minimal wiring.
- Once powered on, the MPU6050 continuously sends raw sensor data to the ESP32, including acceleration along the X, Y, and Z axes, as well as gyroscope readings. These values are sampled at regular intervals, and a software algorithm running on the ESP32 calculates the magnitude of acceleration to determine if abnormal motion has occurred. A sudden drop in acceleration followed by a rapid spike is indicative of a potential fall event.
- The collected sensor data can also be displayed on the OLED screen in real time for monitoring and debugging purposes. For each sensor reading cycle, the ESP32 processes and logs the values locally, which allows for performance tuning and threshold adjustment during testing. This modular sensor integration ensures a reliable data acquisition process and serves as the foundation for accurate fall detection in the wearable system.

1.2.2 Automated Irrigation and Cloud-Based Monitoring:

- Automated irrigation systems coupled with cloud-based monitoring offer a modern solution to enhance agricultural efficiency and sustainability. These systems utilize embedded sensors such as soil moisture detectors, temperature sensors, and humidity modules to continuously monitor field conditions. A microcontroller, such as an ESP32, processes this data and autonomously activates irrigation systems (e.g., water pumps or valves) when soil moisture drops below a predefined threshold. This automation minimizes water waste and ensures optimal soil hydration based on real-time environmental conditions.
- The integration of cloud platforms allows the sensor data to be uploaded wirelessly via Wi-Fi or cellular connectivity. Farmers can access this data remotely through mobile apps or web dashboards, enabling real-time decision-making and historical analysis. Alerts, trends, and irrigation logs can also be viewed through these platforms, offering greater control and insight over crop management. The system reduces manual labor, optimizes water usage, and enhances crop productivity while allowing for scalable and data-driven farming practices.

CHAPTER 2

LITERATURE SURVEY

Over the past decade, numerous studies and prototypes have been developed to address fall detection using wearable technologies. Most conventional systems rely on either threshold-based algorithms or machine learning approaches using data from inertial measurement units (IMUs), such as accelerometers and gyroscopes.

In their work, Bourke et al. (2008) implemented a fall detection system using a single tri-axial accelerometer worn on the waist. Their method relied on identifying peaks in acceleration magnitude, demonstrating a high sensitivity to falls but suffering from false positives in vigorous activities. Similarly, Kangas et al. (2009) investigated multiple sensor placements and found that the combination of acceleration and angular velocity improves accuracy in identifying true fall events.

Recent advancements leverage microcontrollers like Arduino and ESP32, offering greater processing power and wireless capabilities for real-time response. Some researchers have integrated Bluetooth or Wi-Fi for alert transmission, while others have explored mobile application integration for broader monitoring and emergency communication.

Threshold-based detection remains a widely adopted technique due to its simplicity and low power consumption, which is ideal for wearable applications. While machine learning models offer improved precision, they require training data and more power-hungry processors, making them less suitable for lightweight, real-time wearable systems without cloud support.

This project builds on existing research by implementing a threshold-based fall detection system using an MPU6050 sensor and an ESP32 controller, optimized for cost-efficiency, portability, and real-time feedback through a buzzer and OLED display.

2.1 Smart Personal Safety Wearable with Fall Detection:

- This paper proposes a compact, low-cost wearable device designed to detect falls in real-time and send immediate Bluetooth alerts to a connected mobile application. The goal is to provide timely assistance to elderly individuals or patients who are prone to accidental falls.
- The system integrates a wearable fall detection sensor with Bluetooth communication to ensure rapid transmission of alerts to caregivers or medical personnel, significantly reducing response time and potential injury severity.
- Key innovations include a lightweight design, high motion-detection accuracy, efficient hardware components, and seamless mobile app integration, which together enhance accessibility and usability for vulnerable users.
- While the project demonstrates promising feasibility in affordability and portability, it does not deeply address power management or long-term usability, which are critical for real-world continuous monitoring applications.
- This wearable solution builds upon existing fall detection frameworks by focusing on real-time alert delivery through Bluetooth technology, providing a foundational platform for future enhancements like machine learning-based fall classification and integration with broader healthcare monitoring systems.

2.2 IoT-Enabled Monitoring System:

- In this study, the authors present a more robust and scalable system using the ESP32 microcontroller, which supports a wider range of sensors and advanced wireless communication capabilities. The paper highlights the deployment of multiple sensors such as MPU6050, OLED DISPLAY all of which are integrated into a real-time monitoring system.
- One of the key strengths of this project is the inclusion of a flow meter to measure water consumption during irrigation cycles. The system is controlled using a mobile app (Blynk), which provides real-time feedback and manual override functionalities.

- Unlike many earlier models, this system is designed for flexibility and user interactivity, allowing the user to monitor, control, and receive alerts from anywhere using a smartphone. The authors also focus on energy-efficient data transmission, making the system more suitable for remote agricultural fields.
- The methodology includes cloud-based data logging and an intuitive dashboard interface. The advanced use of ESP32 in this system allows better performance, lower power consumption, and higher sensor compatibility. The paper significantly influenced the design of our current system, especially in terms of mobile app integration, sensor expansion, and automation logic.

2.2.1 CareFall: A Comprehensive Framework for Fall Detection Based on Wearable Devices:

- Pham et al. (2023) proposed CareFall, a framework that leverages wearable devices to accurately detect falls in real-time. The system utilizes multiple sensors embedded in wearable devices to monitor motion and posture changes. It employs advanced algorithms that analyze sensor data to distinguish falls from normal activities. Upon detecting a fall, CareFall sends immediate alerts to caregivers through a connected mobile application, facilitating quick response and assistance. The study highlights the importance of precise fall detection in improving elderly care and minimizing injury risks.

2.2.2 Smart Application for Fall Detection Using Wearable ECG and Accelerometer Sensors:

- Yousuf et al. (2022) developed a smart fall detection system integrating ECG and accelerometer sensors within a wearable device. Their system not only detects the abrupt motion typical of a fall but also monitors the physiological signals like heart rate to validate the event's seriousness. The data collected from these sensors is processed in real-time, and if a fall is confirmed, the system triggers an alert notification to emergency contacts. This multi-sensor fusion approach enhances accuracy and reduces false alarms, making it suitable for elderly or at-risk individuals.

2.2.3 BlockTheFall: Wearable Device-Based Fall Detection Framework Powered by Machine Learning and Blockchai:

- Kumar et al. (2023) introduced BlockTheFall, a novel fall detection framework that combines machine learning with blockchain technology. The wearable device collects movement data which is analyzed using machine learning models to detect falls accurately. Unlike conventional systems, BlockTheFall ensures secure and immutable transmission of fall alerts and user data via blockchain, thereby addressing privacy and data integrity concerns. This dual approach of intelligent detection and secure data handling represents a significant innovation in personal safety wearables.

CHAPTER 3

EXISTING SYSTEM

Fall detection is an essential feature in assistive technologies for elderly individuals, patients recovering from surgery, and individuals with physical impairments. Traditional fall detection systems primarily depend on manual alert methods like panic buttons, wired alarms, or calling services. These solutions require active participation from the user, which is not always possible during an actual fall scenario, especially if the individual becomes unconscious or physically immobilized. Thus, manual systems often fail to deliver timely assistance, which can lead to serious health risks.

In more advanced settings, environmental sensor-based systems have been deployed using infrared motion detectors, pressure-sensitive floor mats, or surveillance cameras to monitor movement. While such systems may provide more automation, they come with significant limitations. These setups are confined to indoor environments, lack mobility, and often raise privacy concerns. Furthermore, the infrastructure and maintenance costs of such systems make them impractical for personal or home use, especially in rural or low-income communities.

Recent developments in wearable technologies have introduced accelerometer and gyroscope-based detection systems. These typically involve smartwatches or belt-worn devices that measure sudden changes in motion and orientation. While this approach improves portability and continuous monitoring, many of these commercial products are costly and require a constant internet connection for cloud-based analysis or alerts. Some also depend heavily on smartphones for operation, reducing independence and reliability in emergencies when the phone may be out of reach or out of charge.

In summary, existing fall detection solutions, whether manual, environmental, or wearable, have limitations that affect their accuracy, affordability, or usability. There is a growing need for a low-cost, compact, and reliable fall detection system that operates independently, offers real-time alerting capabilities, and ensures both indoor and limited outdoor usability. The proposed Smart Wearable Fall Detection System aims to fill these gaps using a cost-effective microcontroller (ESP32), an inertial sensor (MPU6050), and simple alert mechanisms like a buzzer and OLED display to provide timely assistance without reliance on external devices or internet connectivity.

3.1 Fall Detection Using Wearable Sensor Devices:

Overview:

One of the most practical approaches in modern personal health monitoring systems is wearable fall detection. These systems use embedded sensors like accelerometers and gyroscopes to monitor a user's motion in real-time. Fall detection is crucial for elderly individuals, patients, or physically vulnerable persons who may not be able to call for help after a sudden fall. The goal of these systems is to immediately detect the fall and alert caregivers or trigger alarms, improving response time and reducing medical risks.

Working:

In most existing wearable systems, motion sensors such as the MPU6050 are integrated into microcontroller platforms like the ESP32. The MPU6050 measures both linear acceleration and rotational movement across three axes. The ESP32 processes this sensor data and detects abrupt acceleration drops or posture changes that indicate a fall. If a fall is confirmed based on threshold-based logic, an alert is generated. In simpler systems, this alert is activated through local outputs like a buzzer or vibration motor. In more advanced systems, the data is sent wirelessly via Wi-Fi or Bluetooth to a smartphone or cloud service. OLED displays are often used to show system status or data output locally on the wearable device.

Benefits:

This type of system offers continuous, real-time monitoring without relying on external inputs. It allows users to carry the system with them, making it ideal for indoor and limited outdoor use. The integration of low-cost hardware components such as the ESP32 and MPU6050 makes the system affordable and compact. Real-time alerts using local indicators like buzzers help ensure immediate awareness, especially when internet access is unavailable. Such systems promote independence and safety for elderly individuals living alone.

Limitations:

Many existing systems are based on simple threshold algorithms, which may cause false positives (e.g., sudden sitting or fast walking mistaken as a fall) or false negatives (minor falls not being detected). Additionally, some solutions depend on continuous power supply or a paired smartphone, reducing usability in rural or offline environments. Another limitation is that basic systems may not store or

analyze data over time, limiting their effectiveness for health trend monitoring or predictive healthcare.

3.2 Vision-Based Fall Detection Systems:

Overview:

Vision-based fall detection systems use cameras and computer vision algorithms to identify human posture and motion patterns. These systems are commonly deployed in hospitals, elderly care homes, or smart homes. Instead of using physical sensors on the body, they monitor a person's activity from a distance using fixed surveillance equipment. This makes them a non-contact solution and suitable for users who are uncomfortable wearing devices.

Working:

The system uses fixed-position cameras, typically ceiling-mounted, to capture continuous video footage. Image processing techniques such as frame differencing, background subtraction, and pose estimation are applied to detect abnormal changes in body orientation. If a person is detected transitioning rapidly from an upright to a horizontal position and remains motionless for a set time, the system interprets it as a fall. These systems often run on local computers or edge devices and may use tools like OpenCV, YOLO, or TensorFlow for detection. Alerts can be sent via SMS, email, or displayed on a central monitoring dashboard.

Benefits:

The main advantage is that users do not have to wear any device. This is especially helpful for people with cognitive challenges or for environments where patient compliance with wearable gadgets is low. Vision-based systems can cover wide areas and monitor multiple individuals at once. They also allow for the recording and playback of fall events, which is useful for post-event analysis.

Limitations:

Despite their advantages, vision-based systems are highly dependent on lighting conditions, camera angle, and environmental clutter. They may not work effectively in dark or occluded environments. Moreover, these systems raise privacy concerns, especially in private living spaces. Installation and maintenance can be expensive due to the need for high-quality cameras and processing hardware. Additionally, real-time detection accuracy may suffer due to motion blur or background interference.

CHAPTER 4

PROPOSED SYSTEM

The proposed system is a wearable fall detection device designed using low-cost and compact components, including the ESP32 microcontroller, MPU6050 sensor, OLED display, and a buzzer. The MPU6050 sensor continuously monitors the user's motion and orientation by capturing acceleration and gyroscopic data across three axes. The ESP32 processes this data and uses a threshold-based algorithm to detect sudden changes in movement that indicate a fall. Once a fall is detected, the ESP32 triggers the buzzer to sound an immediate alert and displays a warning message on the OLED screen, providing both audible and visual feedback.

This system offers a reliable, real-time solution for individuals who may be at risk of falling, such as elderly people or patients with mobility issues. Unlike existing systems that rely on smartphones or internet connectivity, the proposed device functions independently, making it suitable for use in both urban and rural environments. Its wearable and lightweight design allows for continuous monitoring without interfering with daily activities. Furthermore, the system can be enhanced in the future to include wireless alerting features using Bluetooth or Wi-Fi, making it scalable and adaptable to more advanced healthcare applications.

Key Features and Enhancements Over Existing Systems:

1. **Low-Cost and Portable Design:**
   - The proposed system is built using affordable and readily available components such as the ESP32, MPU6050, OLED display, and buzzer.
   - The ESP32 provides high processing capability, built-in Bluetooth and Wi-Fi, and low power consumption, all within a compact module.
   - Unlike camera-based or cloud-dependent systems, this wearable device can operate offline, making it ideal for rural and low-resource environments.
2. **Real-Time Local Alerting Mechanism:**
   - The device detects a fall instantly and activates an onboard buzzer for an audible alert.
   - The OLED screen provides a clear visual warning message, ensuring the user and nearby individuals are notified immediately.
   - This immediate feedback ensures fast local response even in areas without internet or smartphone access.

3. Wearable and User-Friendly:

- The system is lightweight and compact, allowing it to be worn comfortably throughout the day.
- Its operation is fully autonomous—no need to press buttons or configure mobile apps.
- Ideal for elderly users or individuals who may not be comfortable with complex devices.

Benefits of the Proposed System:

- **Improved Safety:** Provides timely alerts when a fall occurs, reducing response time and potentially preventing further injury.
- **Increased Independence:** Enables elderly users or patients to move freely with confidence that help will be alerted in case of a fall.
- **No Network Dependency:** Operates entirely offline, which is essential in locations with limited or no connectivity.
- **Highly Adaptable:** The design allows for future upgrades such as Bluetooth-based caregiver alerts or integration with health monitoring systems.
- **Affordable and Scalable:** Suitable for deployment in homes, clinics, or assisted living centers with minimal cost and setup.

CHAPTER 5

SYSTEM ARCHITECTURE

The system architecture of the Smart Personal Safety Wearable with Fall Detection integrates a microcontroller-based hardware unit with essential components including an MPU6050 motion sensor, a buzzer for audible alerts, and an OLED display for real-time feedback. The ESP32-WROOM module serves as the core processing and communication unit, continuously reading motion data from the MPU6050 via I2C and analyzing it using a threshold-based fall detection algorithm. Upon detecting a fall, the ESP32 triggers an audible alert through the buzzer and simultaneously communicates via Bluetooth with a connected mobile application to notify caregivers. This compact, low-power design ensures portability and reliability, enabling real-time monitoring and immediate response for users at risk of falls, such as elderly individuals or patients with mobility challenges.

System Descriptions:

1. Sensor Layer (Data Acquisition Layer):

- MPU6050: A 6-axis motion sensor combining a 3-axis accelerometer and a 3-axis gyroscope, it continuously monitors the user's motion and orientation.
- It detects sudden changes in acceleration and angular velocity that may indicate a fall.

2.Control Unit (Processing Layer):

- The ESP32-WROOM module receives real-time motion data from the MPU6050 sensor via I2C communication.
- It processes the data using a threshold-based algorithm to detect unusual motion patterns that resemble a fall.
- When a fall is detected, the ESP32 activates alert mechanisms and manages communication protocols.

3. Communication Layer (Connectivity Layer:

- The built-in Wi-Fi and Bluetooth capabilities of ESP32 allow wireless communication.
- Bluetooth can be used to send alerts to a paired mobile device, while Wi-Fi can connect the device to a cloud or app service for remote monitoring.

4. User Interface (Application Layer):

- An OLED Display provides real-time feedback, including sensor readings and fall status.
- A mobile application (if integrated) may display alerts and allow the user to report or dismiss events.
- Auditory feedback via the buzzer helps notify users and caregivers during critical events.

5. Actuation Layer:

- When a fall is detected, the ESP32 triggers a buzzer to alert nearby individuals.
- The system may also send signals via Bluetooth/Wi-Fi for remote notification to a caregiver.

System Workflow :

1. The MPU6050 sensor constantly monitors the user's motion.
2. Data is sent to the ESP32, which analyzes it in real time.
3. If a fall-like pattern is detected, the buzzer is activated for immediate alert.
4. A message or signal is sent to a connected device (e.g., mobile phone) for remote notification.
5. The OLED display shows system status and fall detection updates.

CHAPTER 6

CIRCUIT AND IMPLEMENTATION

The circuit design for the Smart Wearable Fall Detection System is centered around the ESP32-WROOM microcontroller, which interfaces with a set of carefully selected components to ensure real-time fall detection and alerting capabilities. The MPU6050 sensor is connected via the I2C bus to monitor motion and orientation data, while a 0.96-inch OLED display provides visual feedback directly from the device. An active buzzer is integrated to produce an audible alert when a fall is detected. All components are powered via the ESP32's onboard 3.3V and GND pins, and communication between modules is achieved through GPIO and I2C protocols. The compact nature of the circuit ensures portability, making it ideal for wearable applications. The complete system was assembled on a breadboard for prototyping and tested using the Arduino IDE environment.

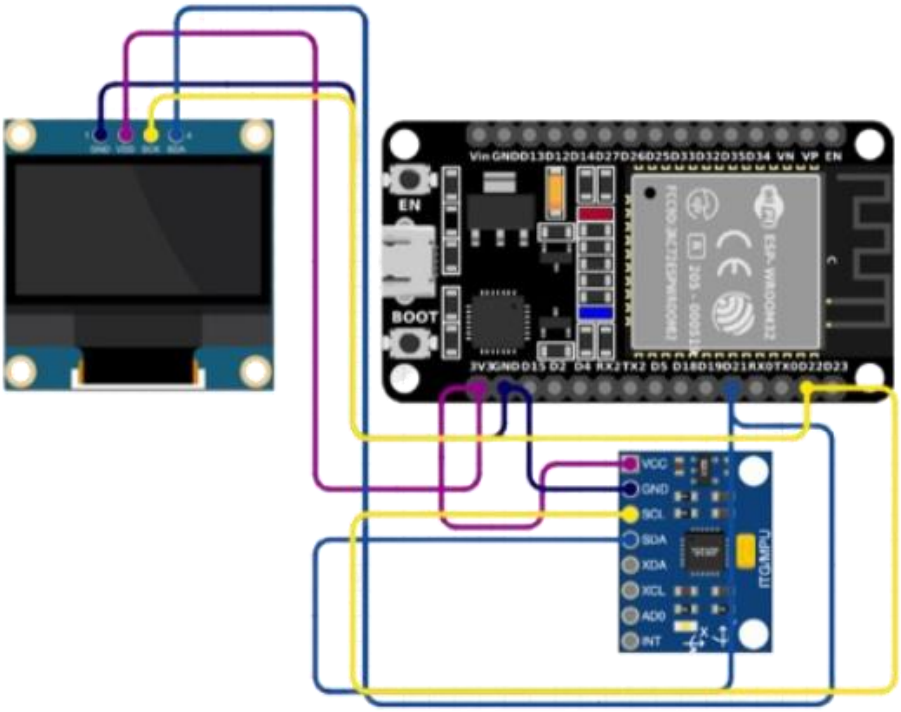

Fig.1 circuit diagram

The core components of the circuit are:

- ESP32-WROOM-32
- MPU6050 Sensor (Gyroscope + Accelerometer)
- OLED Display (0.96" I2C)
- Active Buzzer (5V)
- Jumper Wires and Breadboard

Components Description:

- **ESP32-WROOM-32:**
  A low-power, Wi-Fi + Bluetooth-enabled microcontroller that serves as the system's processing and communication unit.
- **MPU6050 Sensor (Gyroscope + Accelerometer):**
  Detects 3-axis acceleration and rotation to monitor user movement and detect sudden falls.
- **OLED Display (0.96" I2C):**
  Provides real-time display of sensor readings and system status to the user.
- **Active Buzzer (5V):**
  Sounds an alert when a fall is detected, helping notify nearby people of an emergency.
- **Jumper Wires and Breadboard:**
  Used to connect and organize the circuit components during prototyping and testing.

CHAPTER 7

RESULT AND DISCUSSION

The Smart Wearable Fall Detection System was successfully designed and implemented using the ESP32-WROOM microcontroller, MPU6050 sensor, OLED display, and an active buzzer. The prototype was assembled on a breadboard and tested under different motion scenarios to evaluate its response to normal activities versus fall-like movements. The MPU6050 sensor provided real-time acceleration and orientation data, which the ESP32 processed using a threshold-based algorithm to identify fall events.

During testing, the system was able to accurately distinguish between typical daily movements and simulated falls by analyzing the magnitude of acceleration and sudden orientation shifts. When a fall was detected, the buzzer was activated to provide immediate audible feedback, and the OLED screen updated the fall status accordingly. These alerts proved to be effective in real-time scenarios, validating the reliability of the fall detection logic implemented on the microcontroller.

Additionally, the OLED display enabled users to view sensor output and system responses in real time, making the device both functional and user-friendly. The modular hardware design allowed for easy debugging and future expansion. The system's ability to operate on USB power made it suitable for continuous prototyping, and its compact layout supports future transition into a wearable form factor.

CHAPTER 8

CONCLUSION AND FUTURE WORK

The developed Smart Wearable Fall Detection System demonstrates a cost-effective, portable, and responsive solution for detecting falls in real time. By integrating a motion sensor, visual and auditory alerts, and wireless connectivity potential via ESP32, the device provides a reliable means of supporting personal safety, particularly for elderly users or individuals at risk. The implementation proves that threshold-based algorithms, when combined with real-time feedback mechanisms, can offer an efficient and practical solution in wearable health monitoring.

FUTURE WORK:

- Integration of IoT Features: The system can be enhanced by enabling Wi-Fi or Bluetooth-based connectivity to a cloud or mobile app, allowing caregivers to receive remote fall alerts and track user safety through a centralized dashboard.
- Power Optimization and Battery Integration: Future versions of the wearable should include a rechargeable Li-Po battery and power-saving modes (e.g., ESP32 deep sleep) to ensure long battery life and support real-world, mobile usage without the need for constant charging.
- Enhanced Fall Detection via Machine Learning: Incorporating machine learning algorithms trained on real motion data can help improve detection accuracy and reduce false positives, making the system smarter and more adaptive to various fall patterns and user behaviors.

APPENDICES

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <Adafruit_MPU6050.h>
#include <Adafruit_Sensor.h>
#include <BluetoothSerial.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
Adafruit_MPU6050 mpu;
BluetoothSerial SerialBT;

float threshold = 15.0;

void setup() {
  Serial.begin(115200);
  Wire.begin();

  // OLED init
  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println("OLED not found");
    while (1);
  }
  display.clearDisplay();
  display.setTextColor(SSD1306_WHITE);
  display.setTextSize(1);
  display.setCursor(0, 0);
  display.println("Initializing...");
  display.display();

  // MPU init
  if (!mpu.begin()) {
    Serial.println("MPU6050 not found!");
    display.clearDisplay();
    display.setCursor(0, 0);
    display.println("MPU6050 ERROR");
    display.display();
    while (1);
  }
```

```
  mpu.setAccelerometerRange(MPU6050_RANGE_8_G);
  mpu.setGyroRange(MPU6050_RANGE_500_DEG);
  mpu.setFilterBandwidth(MPU6050_BAND_21_HZ);

  // Bluetooth init
  SerialBT.begin("FallDetector");
  Serial.println("Bluetooth started.");

  display.clearDisplay();
  display.setCursor(0, 0);
  display.println("Ready!");
  display.println("Threshold: 15");
  display.display();
}

void loop() {
  sensors_event_t a, g, temp;
  mpu.getEvent(&a, &g, &temp);

  float ax = a.acceleration.x;
  float ay = a.acceleration.y;
  float az = a.acceleration.z;

  float totalAccel = sqrt(ax * ax + ay * ay + az * az);

  Serial.print("Accel: ");
  Serial.println(totalAccel);

  display.clearDisplay();
  display.setCursor(0, 0);
  display.setTextSize(1);
  display.print("Accel: ");
  display.println(totalAccel);

  if (totalAccel > threshold) {
    Serial.println("Fall detected!");
    SerialBT.println("🚨 Fall detected!");
    display.setCursor(0, 20);
    display.setTextSize(2);
    display.println("FALL!");
  } else {
    display.setCursor(0, 20);
    display.setTextSize(2);
```

```
    display.println("SAFE");
  }

  display.display();
  delay(500);
}
```

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